Malley diabase dykes of the Slave craton, Canadian Shield: U–Pb age, paleomagnetism, and implications for continental reconstructions in the early Paleoproterozoic[1]

Kenneth L. Buchan, Anthony N. LeCheminant, and Otto van Breemen

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Abstract: The NE-trending Malley dyke swarm, dated herein at 2231 ± 2 Ma (U–Pb baddeleyite), extends from the central Slave craton to the vicinity of the Kilohigok basin, and may continue farther to the northeast as the geochemically similar Brichta dyke swarm, having been offset sinistrally along the prominent Bathurst fault. It carries a characteristic high unblocking temperature paleomagnetic component of single polarity directed up SE (mean direction: $D = 138.3°$, $I = –53.8°$), with corresponding paleopole at 50.8°S, 50.0°W. Lower unblocking temperature components, in some cases directed down SE, similar to ca 1.75 Ga post-Hudsonian overprints, are easily removed using combined alternating field (AF) thermal demagnetization, but difficult to remove using AF cleaning alone. The characteristic remanence has not been demonstrated primary, but is significantly older than 2.03 Ga, the age of Lac de Gras dykes, based on a baked contact test at a Lac de Gras – Malley dyke intersection. In addition, an E- to ESE-trending dyke carries a down WNW remanence, typical of 2.19 Ga Dogrib dykes near Yellowknife, suggesting that regional overprinting has not affected the study area since Dogrib emplacement, and that the Malley remanence was acquired at or shortly after Malley emplacement. Comparing Malley and Lac de Gras paleopoles with the 2.22–2.00 Ga Superior craton apparent polar wander path indicates that the two cratons were (*i*) not in their present relative orientation at 2.23 or 2.03 Ga, and (*ii*) likely not drifting in close proximity to one another as parts of a single (super)continent throughout the 2.23–2.03 Ga interval.

Résumé : L'essaim de dykes Malley, à tendance nord-est, est daté à 2231 ± 2 Ma (U–Pb sur baddeleyite); il s'étend du craton central des Esclaves jusqu'aux environs du bassin de Kilohigok. Il pourrait s'étendre plus loin vers le nord-est en tant que l'essaim de dykes Brichta, dont la géochimie est semblable et qui a été décalé vers la gauche le long de l'importante faille Bathurst. Cet essaim Malley comporte une composante paléomagnétique de température de déblocage élevée caractéristique, à polarité simple dirigée vers le haut au sud-est ($D = 138{,}3°$, $I = –53{,}8°$), avec un paléopôle correspondant à 50,8° Sud, 50,0° Ouest. Les composantes de température de déblocage inférieure, dans certains cas dirigées vers le bas au sud-est, semblables aux surimpressions post-hudsoniennes à environ 1,75 Ga, sont facilement enlevées en utilisant une démagnétisation thermique / à champ alterné, mais elles sont difficiles à enlever en utilisant uniquement une démagnétisation à champ alterné. La rémanence caractéristique n'a pas été démontrée mais elle est significativement plus ancienne que 2,03 Ga, l'âge des dykes Lac de Gras, selon des tests de contact à une intersection des dykes Lac de Gras-Malley. De plus, un dyke à tendance E à ESE comporte une rémanence ONO vers le bas, typique des dykes Dogrib (2,19 Ga) à proximité de Yellowknife, suggérant que la surimpression régionale n'ait pas touché le secteur à l'étude depuis la mise en place des dykes Dogrib et que la rémanence Malley ait été acquise en même temps ou peu après la mise en place des dykes Malley. Une comparaison des paléopôles Malley et Lac de Gras avec la dérive polaire apparente du craton du Supérieur (2,22–2,00 Ga) indique que les deux cratons (*i*) n'avaient pas leur orientation relative actuelle il y a 2,23 ou 2,03 Ga et (*ii*) ne dérivaient probablement pas près l'un de l'autre en tant que partie d'un seul (super)continent durant tout l'intervalle 2,23–2,03 Ga.

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K.L. Buchan, A.N. LeCheminant,* and O. van Breemen. Geological Survey of Canada, 601 Booth Street, Ottawa, ON K1A 0E8, Canada.

Corresponding author: Kenneth L. Buchan (e-mail: kbuchan@nrcan.gc.ca).

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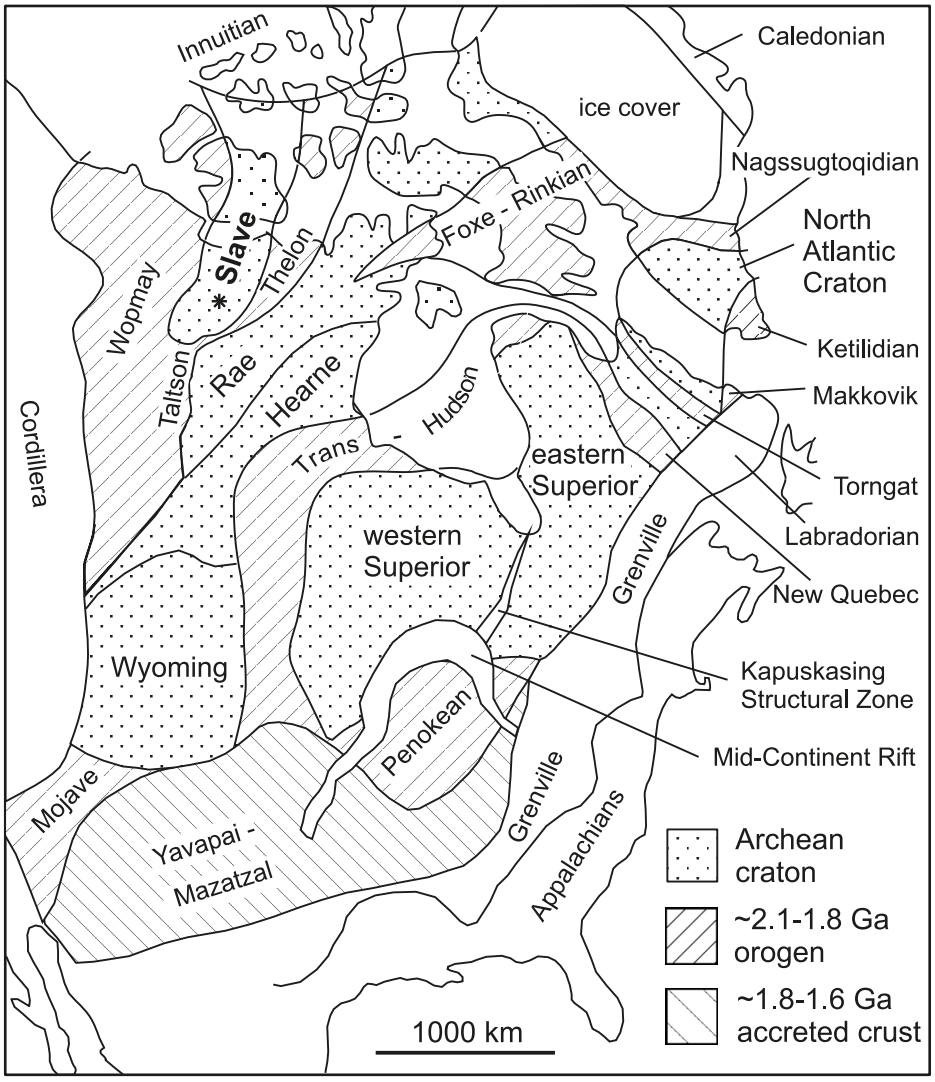
*Present address: 5592 Van Vliet Road, Manotick, ON K4M 1J4, Canada.

Introduction

Thirty to forty Archean cratons or cratonic fragments are scattered widely on all present-day continents. Given that many have Paleoproterozoic rifted margins, it is likely that they were once grouped into one or more (super)continents, which broke apart during the Paleoproterozoic and later amalgamated to form supercontinents such as Nuna (e.g., Bleeker 2003 and references therein). Paleomagnetism provides the only quantitative means to chart the drift of these Archean cratons during the late Archean and Proterozoic, and to test reconstructions of these now-dispersed fragments.

Over the past few years, a sequence of key paleomagnetic poles, those that are both well defined and precisely dated

Fig. 1. Tectonic map of North America and Greenland (modified after Hoffman 1988; Buchan et al. 2009), with Greenland in its pre-Mesozoic position relative to North America. The star indicates the location of the sampling area in the Slave craton. The Superior craton is divided into eastern and western portions across the Kapuskasing Structural Zone (see discussion in text).



(Buchan 2007*b*), were determined for Paleoproterozoic mafic dykes and sills in the Superior craton of Laurentia (Fig. 1) (summary in Buchan et al. 2007*a*). Taken together, these poles constitute the first reliable Paleoproterozoic apparent polar wander path (APWP) established for any of the Archean cratons, and a reference path against which paleopoles from other cratons can be compared to test reconstructions.

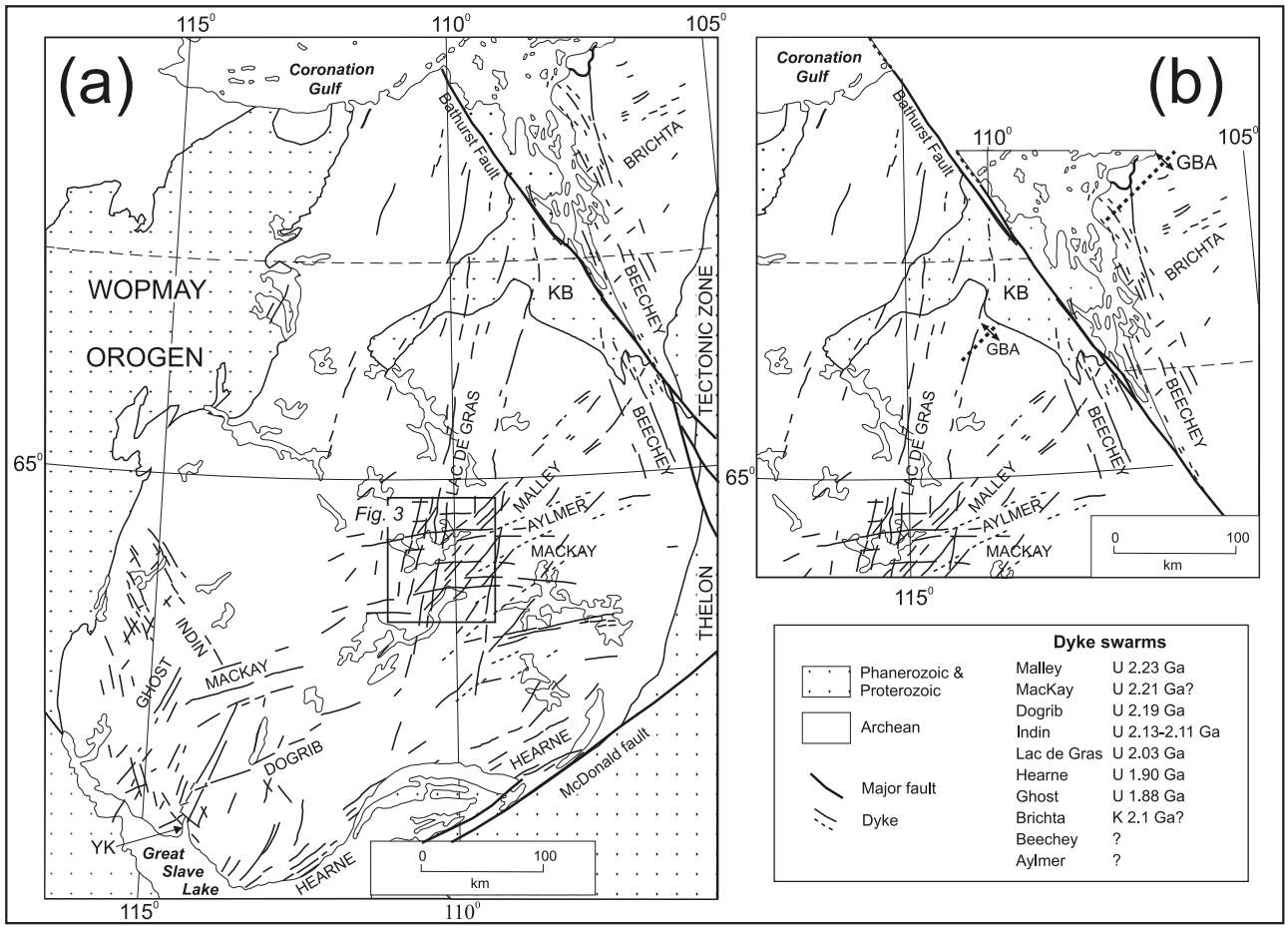
Work is currently underway (Buchan et al. 2007*b*, 2010*b*; Mitchell et al. 2009) to obtain well-defined paleopoles for precisely dated mafic intrusions of the southern Slave craton. The first key Paleoproterozoic pole for the Slave craton was reported from the 2.03 Ga Lac de Gras dyke swarm (Buchan et al. 2009). In this study, we report a precise age of 2231 ± 2 Ma and well-defined paleopole for the Malley dyke swarm, the oldest Paleoproterozoic dyke swarm yet identified in the Slave craton. Although there is no test to demonstrate that the paleopole is primary, it must predate the emplacement of 2.03 Ga Lac de Gras dykes and is likely older than the 2.19 Ga (LeCheminant et al. 1997) Dogrib dykes. The paleopole for the Malley dyke swarm is of particular interest because, if primary, it can be compared with key poles for the slightly younger ca. 2.22 Ga Nipissing sills and Senneterre dykes of the Superior APWP to better understand the geographic relationship between the two cratons in the early Paleoproterozoic.

Geological setting

Laurentia (Fig. 1) comprises several Archean cratonic blocks that are thought to have amalgamated by the late Paleoproterozoic (Hoffman 1988). The geographic relationship between the Slave craton and other Archean cratons prior to amalgamation is uncertain (Bleeker 2003). However, a comparison of primary paleomagnetic poles from the 2.03 Ga Lac de Gras dyke swarm of the central Slave craton and 2.07 Ga Lac Esprit dykes and 2.00 Ga Minto dykes of the Superior Province has confirmed that the two cratons were not in their present relative orientation when these dyke swarms were emplaced (Buchan et al. 2009).

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Fig. 2. Paleoproterozoic diabase dyke swarms of the Slave craton. References for dyke swarms and their ages are summarized in Buchan et al. (2010*a*). (*a*) Present distribution of swarms (simplified from Buchan and Ernst 2004, Buchan et al. 2010*a*). (*b*) Reconstruction of eastern Slave craton based on correction for 115 km of left-lateral offset along the Bathurst fault (Tirrul and Grotzinger 1990), as discussed in the text. Study area of Fig. 3 is enclosed in a rectangle. GBA, Gordon Bay Arch; KB, Kilohigok basin; YK, city of Yellowknife.



The Slave craton is cut by numerous Paleoproterozoic dyke swarms (Fig. 2*a*), many of which may be related to breakup along its margins (LeCheminant et al. 1996). The most prominent swarms in the study area are the NE-trending Malley swarm of the present study, the ca. 2.21 Ga E-trending MacKay (or "X") swarm, whose preliminary age is reported by LeCheminant and van Breemen (1994), and the 2.03 Ga N-trending Lac de Gras swarm. ENE-trending 2.19 Ga Dogrib dykes occur to the southwest near Yellowknife (labelled YK in Fig. 2*a*) and extend to within 75 km of the study area (Fig. 2*a*). Undated Aylmer dykes (Stubley 2005; Buchan et al. 2010*a*) occur mainly to the south and east of the sampling area. They have an ENE trend, intermediate between Malley and MacKay dyke trends. Undated NNW-trending Beechey dykes are located near the Bathurst fault (Fig. 2*a*). Although some Beechey dykes occur in the vicinity of Malley dykes, a crosscutting relationship between the two swarms has not been reported. Poorly dated ENE- to NE-trending Brichta dykes (Bostock et al. 1963; Buchan et al. 2010*a*) occur northeast of the Bathurst fault (Fig. 2*a*). In addition, three sets of Mesoproterozoic dykes are known in the region (Buchan et al. 2010*a*); the dense NW-trending 1.27 Ga Mackenzie swarm (LeCheminant and Heaman 1989) that crosscuts all of the older units, the WNW-trending ca. 1.27 Ga (from paleomagnetic correlation by K. Buchan in LeCheminant 1994) "305" swarm, and the 1.17 Ga (LeCheminant et al. 2007) E-trending Thistlethwaite (or Munn) dyke immediately south of the study area.

The Malley dyke swarm, the main focus of this paper, has been identified from both geological mapping and aeromagnetic data (e.g., Frith 1982*a*, 1987; Fahrig and West 1986; LeCheminant 1994; Lambert 2005). It does not appear to extend a great distance south and east of the study area (Fig. 2*a*). However, it extends to the northeast to within a few kilometres of the Bathurst fault (Fig. 2*a*; Frith 1982*a*; Henderson et al. 1990). Malley dykes have not been observed to cut Goulburn Group sedimentary rocks of the Kilohigok basin (Frith 1987, p. 36). The swarm may be related to breakup along the eastern margin of the Slave craton (LeCheminant et al. 1996), when the craton separated from an, as yet unidentified, (super)continent that predated Laurentia. In the study area, metamorphic grade is generally low. However, the metamorphic grade of Malley and MacKay diabase dykes increases systematically within a few tens of kilometres of the Thelon tectonic zone (fig. 3 of Henderson et al. 1990). Cooling through a temperature of 300 ± 50 °C occurred in this area at ca. 1.74 Ga, based on Rb–Sr biotite ages (fig. 5 of Henderson et al. 1990).

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Previous paleomagnetic studies

Paleomagnetic data have been reported for several dyke swarms in the southern and central Slave craton. Lac de Gras dykes from the study area carry a primary remanence, as noted earlier in the text. The age of the remanence is demonstrated by a positive baked contact test (Everitt and Clegg 1962; Buchan 2007*a*), where a Lac de Gras dyke crosscuts a Malley dyke (see discussion later in the text; Buchan et al. 2009). Paleomagnetic results have not been reported from other dykes of the Malley swarm.

A pair of closely spaced ENE-trending Dogrib dykes in the Yellowknife area (Fig. 2*a*) carry a stable down WNW magnetization (McGlynn and Irving 1975). East-trending MacKay (or X) dykes north of Yellowknife, and in the present study area, carry more complicated remanences (McGlynn and Irving 1975; Fahrig et al. 1984), with directions that Fahrig et al. (1984) interpreted to be a superposition of an up SE, possible primary, remanence that could not be adequately isolated, and a down SE, probably secondary, component. Field tests were not carried out in these studies to establish the ages of the Dogrib and MacKay remanences. Later, Schwarz et al. (1985) attempted three baked contact tests (their profiles 07, 10, and 11), where Dogrib dykes cut Archean volcanics and intrusive rocks in the Yellowknife area. However, their results were inconclusive.

McGlynn and Irving (1975) reported a paleopole for NE- and NW-trending "Indin" dykes over a wide area of the southwestern Slave craton. This pole is no longer considered reliable because the NE- and NW-trending dykes have recently been shown to differ in age by >130 million years, with the NW dykes being dated at ca. 2.13–2.10 Ga (Davis and Bleeker 2007; W. Bleeker, M. Hamilton, and U. Söderlund, personal communication, 2011) and the NE set, renamed the Ghost dykes, dated at ca. 1.88 Ga (Atkinson 2004; Davis and Bleeker 2007; Bleeker et al. 2008). However, a NW-trending Indin dyke at Yellowknife yielded an up W direction and a positive baked contact test (profile 9 of Schwarz et al. 1985), demonstrating that the remanence at this site is primary and dates from the time of dyke emplacement. The fact that the Indin remanence at this site is primary, and that its direction differs by ~90° from that of nearby Dogrib dykes, indicates that there has not been a regional overprint since Indin dyke emplacement and that the Dogrib remanence substantially predates Indin dyke intrusion. Ar–Ar biotite and muscovite studies of Archean rocks near Yellowknife (Bethune et al. 1999) suggest that regional temperatures have not exceeded ~300 °C since ca. 2.4 Ga. This is consistent with the primary nature of the Indin dyke remanence at Yellowknife, based on paleomagnetism, and suggests that the Dogrib dyke remanence is also likely primary.

Petrography and geochemistry

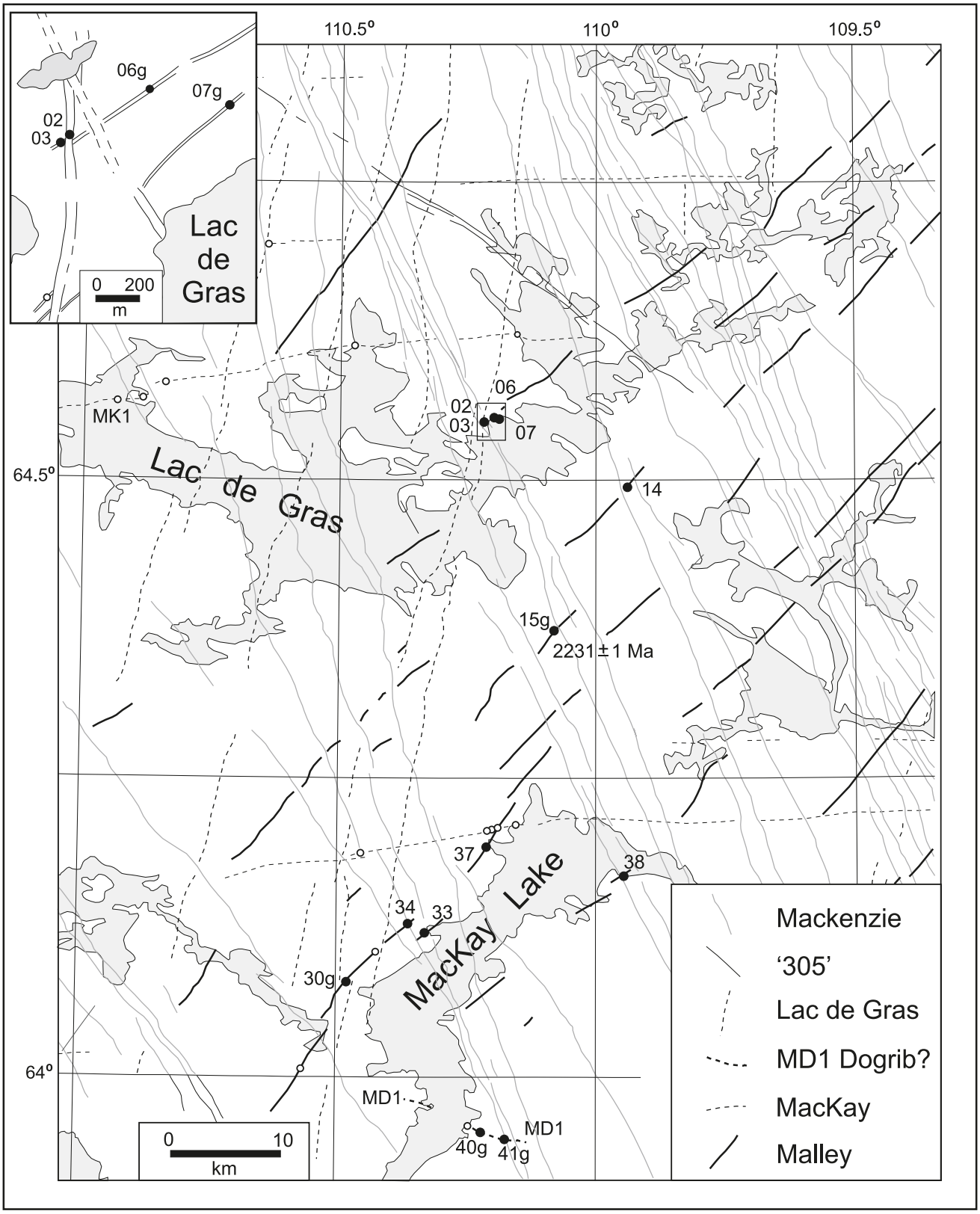
Malley diabase dykes in the study area (Fig. 3) have well-preserved primary igneous textures and well-defined chilled margins. The dated dyke at site 15 southeast of Lac de Gras has a width of 40–45 m and is one of the widest dykes in the region. Otherwise, it is typical of other dykes in the swarm. In thin section, weak alteration is indicated by saussuritized plagioclase, chlorite partly replacing biotite, clusters of epidote crystals, titanite replacement of Fe–Ti oxides, and zircon fringes on primary zirconium-bearing oxides. The dated gabbro contains partly saussuritized plagioclase laths poikilitically enclosed in clinopyroxene. Zoned amphibole rims on clinopyroxene are extensively developed. Blocky Fe–Ti oxide grains are characterized by widely spaced lamellae of ilmenite in titanomagnetite, with variable titanite alteration. Quartz–alkali feldspar-rich interstices contain minor Fe-rich biotite, chlorite, epidote, and trace amounts of apatite, allanite, ilmenite, chalcopyrite, sphalerite, baddeleyite, zirconolite, and zircon. Baddeleyite crystals are generally free of alteration, whereas nearby zirconolite grains are fringed by zircon alteration. A few zirconolite crystals contain inclusions of much smaller baddeleyite. Outer zones of euhedral zircon crystals display patchy alteration associated with tiny specks of thorite.

An E- to ESE-trending, 40 m wide dyke (labelled MD1 in Fig. 3), which has been mapped for >10 km in the vicinity of MacKay Lake, has distinctive geochemistry and a paleomagnetic remanence similar to that of Dogrib dykes (see later in the text). A characteristic feature of this dyke is the presence of sparsely distributed plagioclase megacrysts (1–2 cm). In thin section, coarse diabase from this dyke contains zoned plagioclase poikilitically enclosed in variably altered clinopyroxene with partial amphibole rims. Small baddeleyite and zirconolite needles occur in the outer fringes of amphibole overgrowths and in interstices. Fe–Ti oxides are dominated by skeletal ilmenite associated with partly altered titanomagnetite containing fine exsolution lamellae overprinted by titanite alteration. Interstices contain quartz–alkali feldspar granophyre, Fe-rich biotite and associated chlorite, apatite needles, baddeleyite, and zirconolite. Baddeleyite blades are fresh with sharp crystal boundaries, in contrast to the variable alteration of zirconolite.

Geochemical data for Malley, MacKay, Brichta, and Dogrib dykes (Ernst and Buchan 2010) are compared in Figs. 4*a*–4*d*. Three analyses for the distinctive MD1 dyke are included. Locations for geochemical sampling of Malley, MacKay, and MD1 dykes in the study area are shown in Fig. 3.

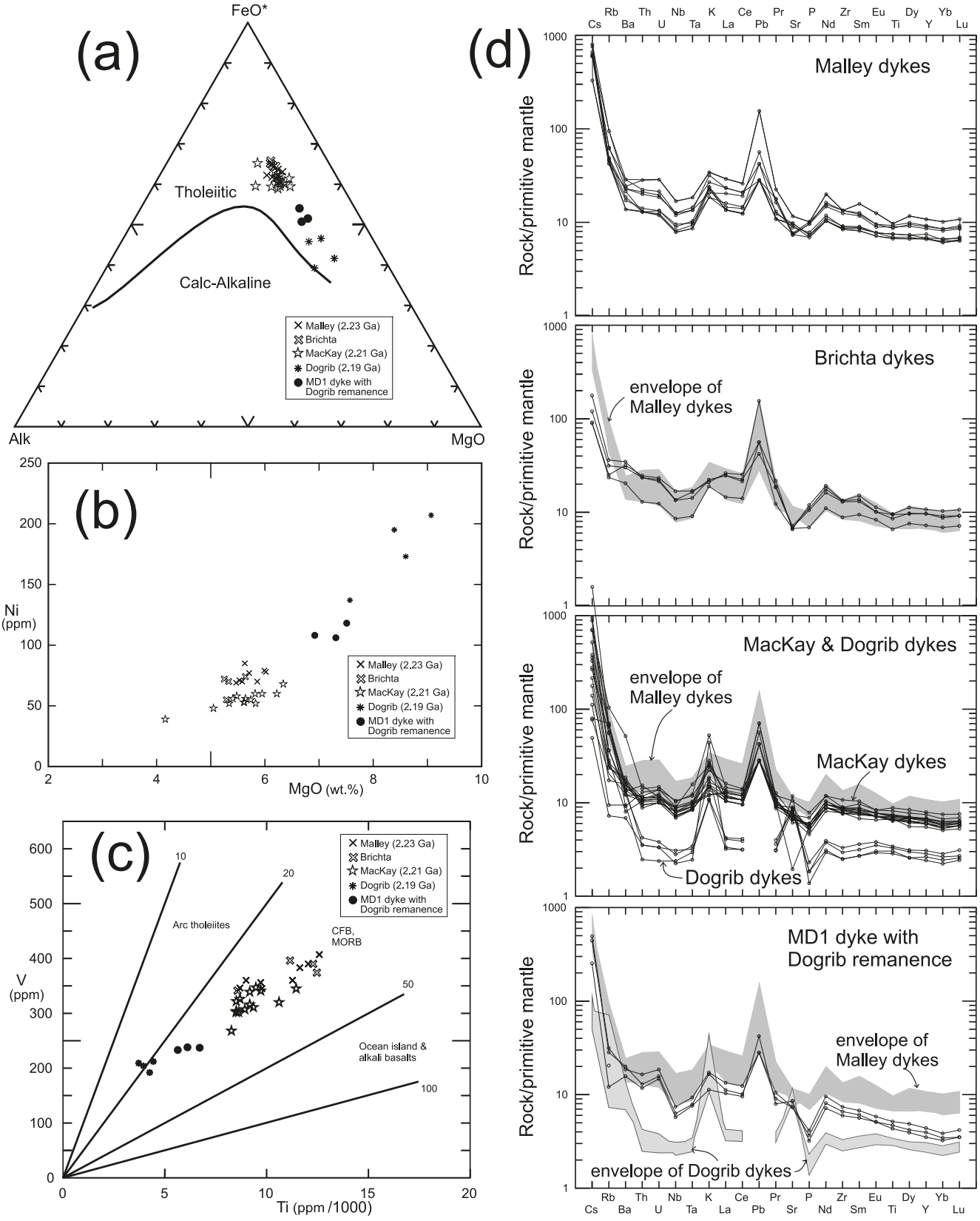
On an alkali – total iron – magnesium (AFM) diagram (Fig. 4*a*), all dyke analyses plot on the tholeiitic side of the tholeiitic–calc-alkaline boundary. The four Dogrib and three MD1 analyses form two clusters with less differentiated compositions than dykes from the other swarms. Dogrib compositions are slightly more primitive than the MD1 dyke, a point confirmed by their higher Ni and MgO contents (Fig. 4*b*), although one Dogrib analysis is similar to MD1 in Ni–MgO. Malley, MacKay, and Brichta compositions are more evolved than either Dogrib or MD1 (Figs. 4*a*–4*d*). These dykes plot in overlapping clusters near the apex of the field boundary on the AFM diagram and have low values of Ni and MgO. On the Ti–V discrimination diagram (Fig. 4*c*), Malley, MacKay, and Brichta dykes plot in the continental flood basalts – mid-ocean ridge basalts (CFB–MORB) field, with no significant distinction between Malley and Brichta dykes. Most MacKay dykes have lower V and Ti values, although there is some overlap with the Malley and Brichta data. Dogrib and MD1 dykes also plot mainly in the CFB–MORB field, but distinct from each another and close to the arc tholeiite field. Spidergrams (Fig. 4*d*) indicate that Malley,

Fig. 3. Diabase dyke swarms (modified after LeCheminant 1994) and sampling sites of the Lac de Gras – MacKay Lake area. Paleomagnetic sites are shown as large closed circles and numbered as in Table 2. A U–Pb baddeleyite age is described in this study from Malley site 15. Geochemical data plotted in Fig. 4 are from paleomagnetic sites labelled "g", as well as unnumbered sites shown as small open circles (geochemical data and site locations are from Ernst and Buchan 2010). MD1 identifies a dyke with a Dogrib paleomagnetic direction and unique geochemistry as discussed in the text. The inset locates sampling sites and shows details of crosscutting relationships for NNW-trending Mackenzie, N-trending Lac de Gras, and NE-trending Malley dykes within the rectangle on the north shore of Lac de Gras.



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Fig. 4. Geochemical diagrams for Malley, Brichta, MacKay, Dogrib, and MD1 dykes generated using the software IGPET version 2008 (Carr 2008). All plotted analyses, with locations, are reported by Ernst and Buchan (2010). (*a*) Alkali – total iron – magnesium (AFM) diagram (boundary between the tholeiitic and calc-alkaline fields after Irvine and Baragar 1971). (*b*) Bivariate plot of Ni versus MgO. (*c*) Ti–V basalt discrimination diagram (after Shervais 1982). CFB, continental flood basalts; MORB, mid-ocean ridge basalts. (*d*) Spider diagrams with data normalized to primitive mantle using values from Sun and McDonough (1989). Note that Pb values for the Dogrib dykes were below the detection limit of 1 ppm.



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MacKay, and Brichta dykes are weakly light rare-earth element (LREE) enriched. MacKay dykes have slightly lower trace element abundances and flatter rare-earth element (REE) profiles than Malley and Brichta dykes. Dogrib dykes have even lower trace element abundances and flat REE patterns in contrast to the other swarms. The MD1 dyke is distinct from all other plotted swarms and has a relatively steep LREE enriched pattern.

In summary, there are strong geochemical similarities between Malley and Brichta dykes, and both these swarms have some overlap in characteristics with MacKay dykes. In addition, the MD1 dyke is unique within the map area. It has a major element composition similar to Dogrib dykes near Yellowknife, but with different trace element and REE characteristics.

U–Pb isotope dilution – thermal ionization mass spectrometry (ID–TIMS) geochronology

Sampling and laboratory procedures

A sample for U–Pb geochronology was collected 15 m from the chilled margin of a 40–45 m wide Malley dyke emplaced in Archean leucogranite and pegmatite at paleomagnetic site 15 (Fig. 3). Processing techniques for the separation of baddeleyite from diabase followed conventional techniques of crushing, Wilfley table heavy mineral concentration, heavy liquids, and magnetic separation. Thermal ionization mass spectrometry, isotope dilution procedures for U and Pb isotopic analysis (ID–TIMS) are described in Parrish et al. (1987). Striated and flat baddeleyite crystals selected for analysis were grouped into fractions based on quality, determined primarily by uniform brown colour and absence of turbidity, fractures, inclusions, and attached grains. Treatment of analytical uncertainties and their propagation is based on procedures outlined by Roddick (1987).

Results

U–Pb data for the Malley dyke at site 15 are presented in Table 1 and displayed on a concordia diagram in Fig. 5. U concentrations range from 251 to 394 ppm. Fractions are 0.3%–0.9% discordant and have $^{207}Pb/^{206}Pb$ ages that are in agreement within the reported uncertainties and yield a weighted average of 2231.4 ± 1.2 Ma (2σ), with a mean square of the weighted deviation (MSWD) of 1.04. The close agreement in $^{207}Pb/^{206}Pb$ ages and the fact that two analyses are nearly concordant suggests that discordance was the result of recent Pb loss. Thus, 2231.4 ± 1.2 Ma is considered to be the best estimate for dyke emplacement and crystallization. A similar age for a Malley dyke is indicated on fig. 11 of French and Heaman (2010), helping to confirm the age of the swarm reported herein.

Paleomagnetism

Sampling and laboratory procedures

Four to six oriented block samples were collected with magnetic and solar compasses at each of 10 NE-trending Malley dyke sites, including the dated site 15 and site 03 baked by a Lac de Gras dyke (Fig. 3). Barring unidentified dyke offshoots or en echelon offsets, the nine unbaked Malley sites likely represent seven individual dykes — five south of Lac de Gras (Fig. 3) and two north of Lac de Gras (inset to Fig. 3). In addition, two sites were sampled along the E- to ESE-trending MD1 dyke that, based on paleomagnetic results described later in the text, may belong to the Dogrib swarm. Baked contact tests were carried out at three of these localities where there are crosscutting younger dykes to help constrain the age of magnetization. At site 34, a NW-trending Mackenzie dyke cuts a Malley dyke. At the location of sites 02 and 03, a N-trending Lac de Gras dyke cuts a Malley dyke (see also Buchan et al. 2009). Finally, at site 41, a Mackenzie dyke cuts the possible "Dogrib" (MD1) dyke. A baked contact test for the Malley dyke remanence was not carried out because no older Paleoproterozoic dykes were observed, and Archean host rocks did not appear to be suitable for such a test.

Cylindrical specimens (2.2 cm in length, 2.5 cm in diameter) were drilled from the oriented block samples, and their remanences measured with a Schonstedt DSM-1 or Molspin Minispin spinner magnetometer. Stepwise alternating field (AF), thermal, or combined AF–thermal demagnetization techniques were used for each specimen. In the combined AF–thermal experiments, stepwise AF cleaning was carried out to fields sufficient to eliminate viscous components (typically 5–20 mT) before stepwise thermal demagnetization was employed. Alternating field cleaning was conducted with a Schonstedt GSD-1 demagnetizer; thermal cleaning with a modified Schonstedt furnace. Experiments were carried out in a shielded room with a residual magnetic field <3000 nT.

Results

Following the elimination of viscous components, all nine unbaked Malley dyke sites are found to carry a high unblocking temperature A magnetization component directed up to the SE (Fig. 6; Table 2), as well as lower unblocking temperature B components, mostly with scattered down directions. The A remanence is consistently of a single polarity, with the exception of one sample at site 06. It is well grouped, with a mean direction of $D = 138.3°$, $I = -53.8°$ ($\alpha_{95} = 5.8°$, $k = 81$), where D and I are declination and inclination, respectively, α_{95} is the radius of the 95% confidence circle, and k is the precision parameter. The corresponding paleopole is located at 50.8°S, 50.0°W (dm = 8.1°, dp = 5.6°, where dm and dp are the semi-axes of the 95% confidence ellipse). As noted earlier in the text, it is likely that seven dykes are represented. This should be a sufficient number to average out much of the secular variation. However, data from additional dykes would help ensure that secular variation is fully averaged out.

The A and B remanence components are easily separated in combined AF–thermal experiments (Fig. 7) because they occupy distinct unblocking temperature ranges. Typically, A unblocking temperatures fall in the 520–580 °C range, indicating that low-Ti titanomagnetite is the remanence carrier. B unblocking temperatures, on the other hand, are usually ≤500 °C, and in many samples ≤350 °C. As a result, a stable A endpoint is obtained in most samples.

Separation of the A and B components is much more difficult using only AF demagnetization because B coercivities can be below, overlapping, or above those of the A component. Examples of AF demagnetization are shown in Fig. 8.

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Table 1. U–Pb isotopic baddeleyite data from NE-trending Malley dyke at site 15.

Fraction[a]	Weight (μg)[b]	U (ppm)	Pb (ppm)[c]	$^{206}Pb/^{204}Pb$[d]	Pb (pg)[e]	$^{208}Pb/^{206}Pb$[f]	$^{207}Pb/^{235}Pb$[f]	$^{206}Pb/^{238}Pb$[f]	$^{207}Pb/^{206}Pb$[f]	$^{206}Pb/^{238}Pb$[g]	$^{207}Pb/^{235}Pb$[g]	$^{207}Pb/^{206}Pb$[g]	Disc[h]
92LAAT35-1, z2899 (64.3747°N, 110.0779°W)													
A, 60 × 40, n = 32	4	251	104	1477	18	0.03	7.945 ±0.12	0.4103 ±0.09	0.14044 ±0.06	2216.1 ±3.5	2224.7 ±2.2	2232.7 ±2.2	0.88
B, 70 × 50, n = 39	9	340	141	1243	69	0.04	7.976 ±0.13	0.4123 ±0.11	0.14029 ±0.06	2225.6 ±4.1	2228.3 ±2.3	2230.8 ±2.1	0.27
C, 70 × 50, n = 47	12	394	163	1416	91	0.03	7.967 ±0.12	0.4119 ±0.09	0.14029 ±0.06	2223.4 ±3.5	2227.3 ±2.1	2230.8 ±2.0	0.39

[a]Nonmagnetic at a 10° side slope, given in degrees on a Frantz isodynamic magnetic separator operating at 1.8 A; sizes of fragments in μm, i.e., 100 × 50; average length and breadth aspects, thickness is generally one-third of the breadth.
[b]Error on weight is ±1 μg.
[c]Radiogenic Pb.
[d]Measured ratio corrected for spike and Pb fractionation of 0.09% ± 0.03% per atomic mass unit (AMU).
[e]Total common Pb in analysis corrected for fractionation and spike.
[f]Corrected for blank Pb and U, common Pb, errors quoted are 1σ in percent.
[g]Age errors quoted are 2σ in Ma.
[h]Discordance in percent along discordia to origin.

At site 37, specimen 37-01-01 carries only the A direction, whereas specimen 37-03-01, exhibits a swing to the A direction, followed by a swing to the SE and down. In contrast, AF–thermal demagnetization of a second specimen (37-03-02) from the latter sample isolates the A component in the high unblocking temperature range (Fig. 7). Specimen 07-02-01 has a rather scattered down SE direction on AF demagnetization (Fig. 8), but a specimen from the same sample (not shown) attains a stable A endpoint on AF–thermal demagnetization.

Although B components are sometimes quite scattered, in many samples they are directed down to the SE, and likely represent post-Hudsonian overprinting at ca. 1.75 Ga. A similar overprint component has been widely recognized in North America (e.g., Irving et al. 2004). It is always of one polarity and, hence, may have been acquired during a magnetic superchron, termed the "Cleaver" superchron because the same magnetization direction is observed in precisely dated 1740 $^{+5}_{-4}$ Ma Cleaver dykes of the Wopmay orogen (Irving et al. 2004).

Paleomagnetic data for the E–ESE-trending MD1 dyke (Fig. 3) at sites 40 and 41 suggest a correlation with the Dogrib swarm at Yellowknife, 200 km southwest of the sampling area, rather than with E-trending MacKay dykes. Samples at site 40 all carry a very stable down WNW remanence (Fig. 6), indistinguishable from directions observed in Dogrib dykes (McGlynn and Irving 1975), but unlike the SE remanence direction of MacKay dykes that has been interpreted as a composite of an older SE up direction and a SE down overprint (Fahrig et al. 1984).

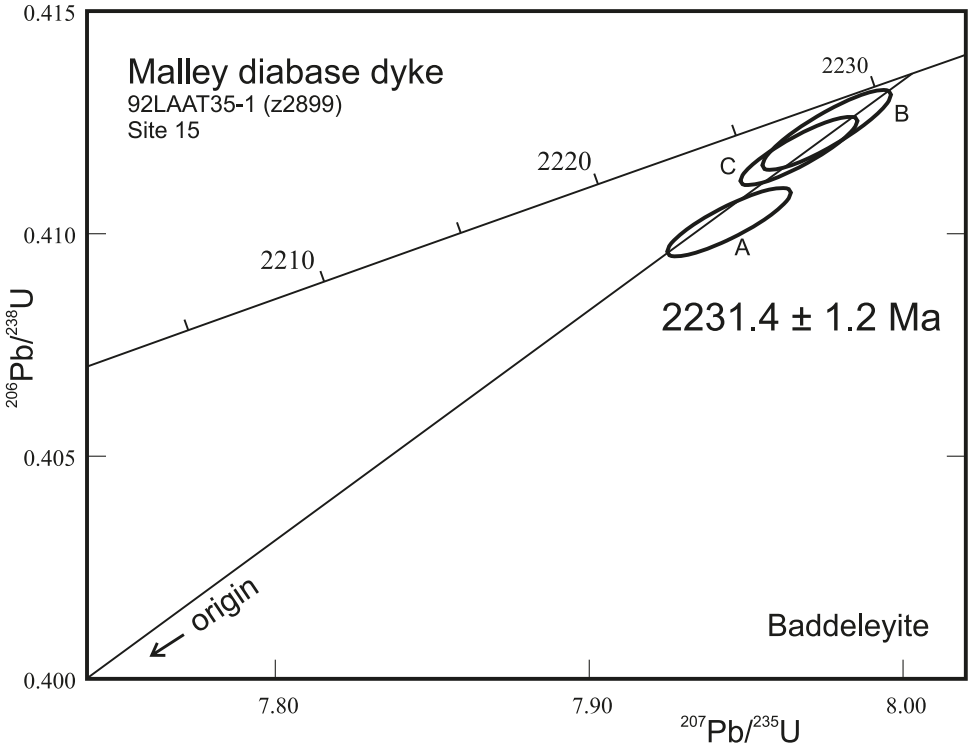
At site 41, the MD1 "Dogrib" dyke is crosscut by a NNW-trending 1.27 Ga Mackenzie dyke. Samples of the MD1 dyke collected close to the Mackenzie dyke carry the same paleomagnetic direction as the Mackenzie dyke. A sample collected far from the Mackenzie dyke carries a single component remanence similar to that observed at site 40, supporting the correlation of this dyke with Dogrib dykes. See further discussion of the results from site 41 later in the text.

Baked contact tests

Positive baked contact tests, conducted where 1.27 Ga Mackenzie and 2.03 Ga Lac de Gras dykes crosscut either Malley or "Dogrib" dykes, demonstrate that the remanence in the Mackenzie and Lac de Gras dykes are primary, the Malley A remanence is substantially older than 2.03 Ga, and the "Dogrib" remanence is substantially older than 1.27 Ga.

At site 34, a NW-trending 1.27 Ga Mackenzie dyke with a width of 17 m crosscuts a Malley dyke (Fig. 9*a*). The Malley dyke is only exposed to within 8.8 m of the Mackenzie dyke. A sample from the Mackenzie dyke carries a remanence similar to that observed in other Mackenzie dykes (e.g., Robertson 1969), whereas unbaked Malley samples collected 40–50 m from the Mackenzie dyke contact carry the Malley A remanence. A Malley sample collected as close as possible (8.8 m) to the Mackenzie contact carries a hybrid remanence, with a direction that swings along the great circle between Mackenzie and Malley directions. This positive baked contact test indicates that the Mackenzie remanence is primary and the Malley A remanences substantially predates emplacement of the Mackenzie dyke.

Fig. 5. U–Pb isotopic ratios plotted on a concordia diagram for baddeleyite fractions from Malley dyke site 15.



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At sites 02 and 03 (Fig. 3, inset), a N-trending Lac de Gras dyke with a width of 30 m crosscuts and magnetically overprints a Malley dyke, whereas the same Malley dyke at site 06, ~500 m to the northeast, carries the Malley A remanence. Detailed paleomagnetic results from this baked contact test are described by Buchan et al. (2009) and illustrated in their fig. 10. This positive test demonstrates that the Lac de Gras remanence is primary and that the Malley dyke A remanence was acquired before 2.03 Ga, the age of the Lac de Gras swarm. Because the Malley A and Lac de Gras remanences have directions that differ by ~70°, the Malley A remanence is likely substantially older than 2.03 Ga.

At site 41, a NNW-trending Mackenzie dyke with a width of ~21 m crosscuts the E- to ESE-trending MD1 "Dogrib" dyke (Fig. 9*b*). A sample from the Mackenzie dyke and three baked "Dogrib" dyke samples (collected at the Mackenzie contact and at 4.0 and 22.9 m from the contact) carry a consistent remanence direction, similar to that observed in other Mackenzie dykes. An unbaked "Dogrib" sample collected 80 m from the Mackenzie contact yields a remanence direction that is similar to that of site 40 (~2.2 km farther west along the same "Dogrib" dyke) and Dogrib dykes at Yellowknife (McGlynn and Irving 1975). This positive baked contact test demonstrates that the MD1 "Dogrib" dyke carries a remanence that significantly predates intrusion of the Mackenzie dyke.

Discussion

Malley dyke swarm

The 2231 ± 2 Ma Malley dyke swarm is the oldest Paleoproterozoic swarm that has been precisely dated in the Slave craton. The consistent paleomagnetic results from Malley dykes in this study indicates that the swarm is at least 40 km wide in the vicinity of Lac de Gras and MacKay lakes. However, based on geological and aeromagnetic mapping of NE-trending dykes, the swarm likely has a width of at least 200 km (Fig. 2*a*; Buchan et al. 2010*a*). As described earlier, Malley dykes can be traced northeast along strike for ~300 km (Fig. 2*a*; Buchan et al. 2010*a*), to within a few kilometres of the Bathurst fault.

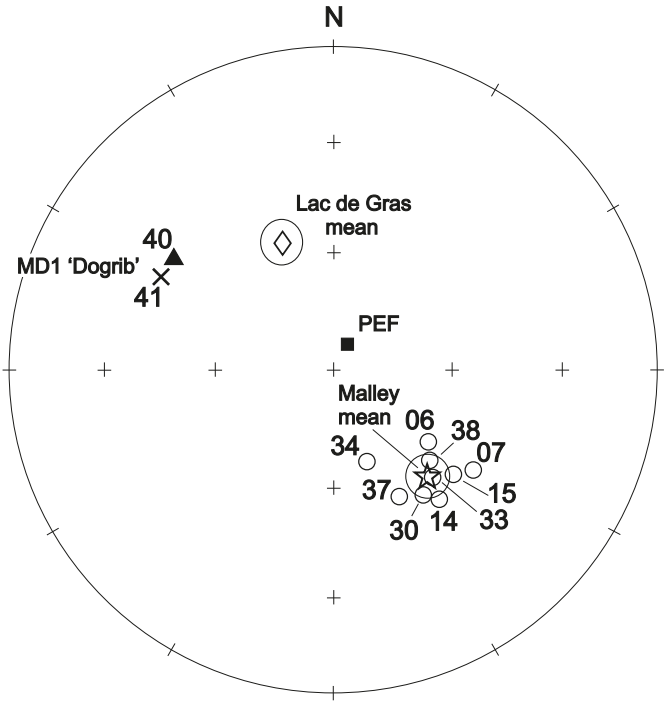
Reconstruction of Malley dyke swarm across Bathurst fault

Malley dykes have not been reported northeast of the Bathurst fault (Fig. 2*a*), along which about 50–140 km of left-lateral displacement is thought to have occurred as a result of indentation of the Slave craton into the Thelon tectonic zone. The 50 km estimate is based on matching aeromagnetic anomalies in the Thelon tectonic zone (Thomas et al. 1976); the 140 km estimate is based on stratigraphic markers in the Goulburn Group (Campbell and Cecile 1979; Tirrul 1985). Tirrul argued for the higher figure, explaining that "to the south the fault system becomes increasingly ductile (Frith 1982*b*), so that discrete large scale offsets record only a small part of the total slip". Subsequently, an intermediate estimate of 115 km was put forward by Tirrul and Grotzinger (1990). A reconstruction based on this estimate aligns the Gordon Bay Arch (Tirrul and Grotzinger 1990; McCormick and Grotzinger 1993), a flexural arch that was active during an early phase of deposition of the Goulburn Group, on either side of the fault.

Poorly dated early Paleoproterozoic Brichta dykes (K–Ar 2.1 Ga; Wanless et al. 1965, p. 44) are located northeast of the Bathurst fault (Fig. 2*a*; Bostock et al. 1963; Thompson et al. 1985, Buchan et al. 2010*a*). They trend ENE to NE,

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Fig. 6. Paleomagnetic directional data for Paleoproterozoic dykes of the Lac de Gras – MacKay Lake sampling area. Data are plotted on an equal-area projection, where open symbols indicate up directions, and closed symbols and a cross indicate down directions. Circles are A site mean directions for NE-trending Malley dykes. The mean of the nine Malley A site directions is shown as a star along with its circle of 95% confidence. The triangle is the paleomagnetic direction from the E- to ESE-trending MD1 "Dogrib" dyke at site 40; the cross is the direction from the single unbaked sample at site 41 on the same dyke. The mean direction for N-trending Lac de Gras dykes (Buchan et al. 2009) is shown as a diamond with its 95% confidence circle.The square indicates the direction of the present Earth's magnetic field (PEF).



and may represent the continuation of the Malley swarm as illustrated in Fig. 2*b*, where we reconstruct the eastern Slave craton to account for the most recent estimate of 115 km offset along the Bathurst fault. Applying 140 km offset shifts the Brichta dykes 25 km further southeast than shown in Fig. 2*b*, but does not alter the preceding interpretation. The geochemical characteristics of Brichta and Malley dykes are similar (as discussed earlier in the text), supporting a correlation between the two swarms. The combined Malley and Brichta swarms have a length of ~500 km and a width of ~200 km.

Reconstruction of Beechey dyke swarm across Bathurst fault

In the current configuration (Fig. 2*a*), undated NNW-trending Beechey dykes occur both southwest and northeast of the Bathurst fault. Southwest of the fault they are found mainly southeast of the Kilohigok basin between latitude 65° and 65.8°N, but have been reported to intrude the base of the Goulburn Group in the southeastern portion of the basin (Frith 1981). Northeast of the fault they are found between ~65.7°N and the Arctic coast. If the Beechey dykes were emplaced before movement occurred on the Bathurst fault, then the reconstruction of Fig. 2*b* eliminates most of the apparent offset between the main portions of the swarm on either side of the fault, and indicates that the swarm originally had a length of at least 200 km and a width of at least 120 km.

Age-correlative magmatic events on other cratonic blocks

Several dyke swarms and sill or volcanic provinces on other Archean cratons have U–Pb ages within 20 million years of the age of the Malley swarm. The 2235 ± 2 Ma Kikkertavak dykes of the Nain Province of the North Atlantic craton (Cadman et al. 1993) and ca. 2240 Ma dykes of the Vestfold Hills in Antarctica (Lanyon et al. 1993) are slightly older than the Malley swarm. On the other hand, ca. 2218–2216 Ma Nipissing sills and Senneterre dykes of the Superior craton (Andrews et al. 1986; Noble and Lightfoot 1992; Buchan et al. 1993), ca. 2220–2206 Ma Koli sills of Karelia (summary in Vuollo et al. 1995), ca. 2220–2209 Ma Kandlamadugu and Somala dykes of the Dharwar craton of India (French and Heaman 2010), 2214 ± 10 Ma BN-1 dykes of the North Atlantic craton in Greenland (Nutman et al. 1995), and ca. 2209 ± 15 Ma Cheela Springs Basalts of the Hamersley basin of the Pilbara craton of Australia (Martin et al. 1998) are a little younger. Of these units, only the Nipissing sills and Senneterre dykes of the Superior craton have a key paleopole (Buchan et al. 1993). A comparison between the Slave and Superior paleomagnetic results are discussed later in the text.

In addition, a primary paleopole has been reported for Ongeluk lavas of the Kaapvaal craton (Evans et al. 1997), which may be roughly coeval with the Malley dykes, based on a Pb–Pb age of 2222 ± 13 Ma (Cornell et al. 1996). However, other authors (e.g., Bau et al. 1999) questioned this age based on an older Pb–Pb age of 2394 ± 26 Ma from overlying Mooidraai dolomite. Therefore, a precise U–Pb age is required to test an Ongeluk–Malley age correlation.

Age of Malley dyke remanence

A field test is not available to demonstrate that the Malley remanence is primary. Nevertheless, it is substantially older than 2.03 Ga, based on the positive baked contact test at the Lac de Gras – Malley intersection. In addition, evidence suggests that it significantly predates 2.12 Ga and likely predates 2.19 Ga, as discussed later in the text.

As described earlier, NE-trending Malley dykes carry a consistent up SE remanence, whereas the E- to ESE-trending MD1 "Dogrib" dyke at sites 40 and 41 is magnetized down WNW (Fig. 6), similar to 2.19 Ga Dogrib dykes near Yellowknife. There are two possible intepretation of these results. Most likely, the difference between the remanences in the Malley and Dogrib dykes indicates that there has been no previous overprinting of the dykes in the study area at or before 2.19 Ga. If this is the case, the "Dogrib" remanence at sites 40 and 41 is primary, and the Malley remanence was acquired significantly before 2.19 Ga, either at the time of or shortly after dyke emplacement at 2.23 Ga.

In the alternative, and less likely interpretation, the agreement of the MD1 "Dogrib" dyke remanence and its polarity with that of Dogrib dykes near Yellowknife and the difference with the remanence direction of Malley dykes are sim-

Table 2. Paleomagnetic results.

Site	Dyke	Trend (°)	Width (m)	Lat. (°N) NAD 83	Long. (°W) NAD 83	N (n_A,n_T)	D (°)	I (°)	k	α_{95} (°)
02[a]	Lac de Gras	015	~30?	64.5508	110.2142	6(3,0)	342.6	–47.1	40	20
03[a]	Malley baked by Lac de Gras	045	11	64.5509	110.2148	4(4,0)	328.8	–51.8	120	8
06[a,b]	Unbaked Malley	045	16	64.5535	110.2064	5(0,4)	126.9	–59.8	56	12
07	Malley	049	—	64.5530	110.1971	4(0,4)	125.6	–45.7	27	18
14	Malley	040	~60	64.4942	109.9342	5(0,4)	140.7	–47.1	98	10/11[c]
15	Malley	~040	40–45	64.3747	110.0779	6(0,5)	131.0	–49.2	141	6
30	Malley	050 (local 035)	25.5	64.0797	110.4748	5(0,4)	144.3	–50.6	68	11
33	Malley	015	45	64.1222	110.3266	6(1,4)	137.4	–52.7	185	6
34-A[d]	Mackenzie	345	17	64.1298	110.3597	1(1,0)	249.1	28.6	—	—
34-B[d]	Malley partly baked by Mackenzie	~060	13.6	64.1298	110.3597	1(—,—)	—	—	—	—
34-C[d]	Malley	060	13.6	64.1298	110.3597	5(0,4)	159.9	–65.3	52	14/16[c]
37	Malley	040?	~25	64.1923	110.2073	5(5,0)	152.5	–53.6	200	5
38	Malley	056	18	64.1694	109.9420	5(1,4)	133.4	–56.3	99	8
40	MD1 "Dogrib"	107	40	63.9555	110.2230	5(5,0)	304.8	39.8	62	10
41-A[e]	Mackenzie	158	~21	63.9509	110.1801	1(1,0)	242.9	40.2	—	—
41-B[e]	MD1 "Dogrib" baked by Mackenzie	E	—	63.9509	110.1801	3(3,0)	241.7	38.2	123	11
41-C[e]	Unbaked MD1 "Dogrib"	E	—	63.9509	110.1801	1(1,0)	298.4	39.3	—	—

Note: Lat. and long. are latitude and longitude, respectively. NAD is North American Datum. N is the number of oriented samples collected at the site. n_A and n_T are the number samples used to calculate the site mean direction that were demagnetized by alternating field (AF) or thermal techniques, respectively. D and I are the mean declination and inclination of the characteristic remanence. k is the precision parameter. α_{95} is the radius of the circle of 95% confidence. The mean direction for the nine stably magnetized Malley dyke sites is $D = 138.3°$, $I = –53.8°$ ($\alpha_{95} = 5.8°$, $k = 81$). The corresponding paleopole is at 50.8° S, 50.0°W (dm = 8.1°, dp = 5.6°).

[a]Sites 02, 03, and 06 constitute a baked contact test for a Lac de Gras dyke cutting a Malley dyke (see text and Buchan et al. 2009).

[b]One of the four stable samples at site 06 is reversely magnetized with a direction of $D = 304.3°$, $I = 72.2°$.

[c]Minimum and maximum values of α_{95}, calculated from remagnetization circle analysis of Bailey and Halls (1984) using three endpoints and one great circle in each of sites 14 and 34-C.

[d]34-A, 34-B, and 34-C constitute the baked contact test for a Mackenzie dyke cutting a Malley dyke (Fig. 9*a*).

[e]41-A, 41-B, and 41-C constitute the baked contact test of a Mackenzie dyke cutting the MD1 dyke (Fig. 9*b*).

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ply coincidental, and a regional dual-polarity overprint in the study area and perhaps in the Yellowknife area has completely erased the primary remanence from the two swarms during or after emplacement of the Dogrib dykes at 2.19 Ga. In this case, it could only be concluded that the Malley (and Dogrib) remanences are significantly older than ca. 2.12 Ga, based on the conclusion reached earlier that the remanence of Dogrib dykes in the Yellowknife area is older than the ca. 2.12 Ga emplacement age of Indin dykes. However, there is no evidence of a regional post 2.19 Ga metamorphic event in the study area or near Yellowknife that could have completely reset high unblocking temperature remanences of the Malley and Dogrib dykes. Therefore, we conclude that this scenario is unlikely, and the Malley remanence was acquired at the time of or shortly after dyke emplacement.

Implications for paleocontinental reconstructions

Assuming that the paleopole for the 2.23 Ga Malley swarm is primary or acquired shortly after dyke emplacement, as seems likely, it is useful to compare it with the primary pole for the crosscutting 2.03 Ma Lac de Gras dykes and with the sequence of key poles that define an APWP for the Superior craton, especially the pole for the 2.22 Ga Nipissing sills and Senneterre dykes, which were intruded shortly after the Malley dykes (Fig. 10). The virtual geomagnetic pole from the well-dated 2.19 Ga Dogrib dykes is also shown for comparison, although it must be used with caution because it is based on only two dykes, does not average out secular variation, and, hence, its ellipse of confidence may greatly underestimate its uncertainty.

It should be noted that relative polarity of the Superior poles is known because the time intervals (≤50 million years) between poles are too short to allow the use of antipoles in the APWP. Similarly, the relative polarity between the Malley and Dogrib poles is known. However, the polarity of Malley and Dogrib poles relative to the Lac de Gras pole is unclear because of the long time gap (~160 million years) between the Dogrib and Lac de Gras poles. Malley and Dogrib poles are located in Fig. 10*a* using the polarity that minimizes their distance from the Lac de Gras pole, and in Fig. 10*b* using the opposite relative polarity option. It should also be noted that the relative polarity of Superior and Slave poles is unknown, so that the Slave craton poles could be located on the opposite side of the globe to their position in Fig. 10. The Slave and Superior poles are plotted on the same side of the globe in Fig. 10 simply for convenience.

The paleopole for 2.23 Ga Malley dykes lies ~15° from the virtual geomagnetic pole for the 2.19 Ga Dogrib dykes

Fig. 7. Examples of paleomagnetic data for Malley dykes using combined AF–thermal demagnetization. Specimen are identified by site (first two digits), sample (third and fourth digits), and specimen (last two digits). Paleomagnetic directions are shown on equal-area plots, with closed (open) symbols indicating down (up) directions. Circles (crosses) on vector diagrams indicate projection onto the horizontal (vertical) plane. Intensity decay plots are shown for the thermal demagnetization portion of the experiments, with intensity (J) normalized to the intensity of the natural remanent magnetization (J_{NRM}).

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(Fig. 10), consistent with the relatively short time interval (~40 million years) between the emplacement of the two swarms. The Malley and Dogrib poles are far from that of the 2.03 Ga Lac de Gras dykes, indicating that substantial drift of the Slave craton occurred between 2.19 and 2.03 Ga, and, hence, that plate tectonic processes were operating during this interval.

Malley dykes are only slightly older than 2.22 Ga Nipissing sills and Senneterre dykes of the Superior craton, which have been shown to carry a primary remanence (e.g., Buchan 1991). The Malley and Nipissing poles are distinct (Fig. 10), indicating that the two cratons were not in their present relative positions at 2.23–2.22 Ga. This is consistent with an earlier conclusion that the two cratons were not in their present relative positions at ca. 2.03 Ga based on a comparison of the Lac de Gras paleopole and 2.07–2.00 Ga poles from the Superior craton (Buchan et al. 2009).

In utilizing paleomagnetic data from two cratons to establish their relative locations, it is best to compare detailed APWPs for the two blocks. Paths with similar shapes indicate the two blocks were drifting in unison on the same tectonic plate; differing shapes indicate that they were on separate plates. However, as yet there is no APWP for the Slave craton. An alternative approach (Buchan et al. 2000), when APWPs are not available, is to compare individual paleopoles of identical age from the two cratons. This is rather difficult in the case of the Slave and Superior data because of the mismatch in age between Dogrib and Lac de Gras poles and poles from the Superior craton. Nevertheless, a preliminary analysis is warranted.

In Fig. 11, the latitudinal positions and azimuthal orientations of the Slave craton at 2.23, 2.19, and 2.03 Ga are compared with the latitude and orientation of the Superior craton as it drifted through the 2.22–2.00 Ga interval. It is important to note that paleolongitude is unconstrained by the paleomagnetic data, so that the relative longitudinal position of Slave and Superior cratons in Fig. 11 is arbitrary. Because relative polarity of the Slave and Superior paleopoles is not known, two polarity options are possible. The Superior craton is arbitrarily placed in one hemisphere, whereas the Slave craton is shown in both hemispheres, with either position permitted by the paleomagnetic data.

It should also be noted that paleomagnetic data indicate that the western portion of the Superior craton (Fig. 1) rotated counterclockwise ~15° relative to the eastern Superior sometime after 2.07 Ga (Halls and Davis 2004; Buchan et al. 2007*a*; Evans and Halls 2010). Therefore, to simplify the analysis which follows, the Slave craton data are only compared with results from the eastern Superior.

If one chooses the polarity options in Fig. 11 that place the Slave and Superior cratons in opposite hemispheres, then the two cratons are at quite different latitudes and could not have been near neighbours throughout the 2.23–2.03 Ga period. If one chooses polarity options with Slave and Superior cratons in the same hemisphere, then Slave craton paleolatitudes at ca. 2.23, 2.19, and 2.03 Ga are roughly similar to those of the Superior craton through the 2.22–2.00 Ga period. However, the *relative* azimuthal orientation of the two cratons differs significantly. For example, the relative orientation at 2.23–2.22 and 2.07–2.00 Ga differs by ~70°–90° (depending on whether the Lac de Gras orientation is compared with that for the somewhat younger Minto or older Lac Esprit dykes). This suggests that, if the two cratons were in close proximity to one another, they were not moving together on the same tectonic plate throughout this period. This interpretation is in agreement with geological evidence that the two cratons were not near neighbours during the early Paleoproterozoic (Bleeker 2003), including their distinct records of mafic magmatic events in this period. Further Slave craton paleopoles are required to test whether the two cratons moved together during some portion of the 2.23–2.22 Ga period.

Because longitude is unconstrained in Fig. 11, it is not immediately obvious whether the Slave and Superior cratons may have drifted during this period at a large distance from one another, but on the same plate, perhaps as distant parts of a single supercontinent. This possibility can be examined by comparing the great-circle distance between pairs of poles from the two cratons that have identical age ranges (Evans and Pisarevsky 2008). Significantly different great-circle distances would indicate that the cratons were on separate plates. Similar great-circle distances are a necessary, but not sufficient, condition for determining that the cratons were on the same plate. In the case of the Slave and Superior data of Fig. 10, such a comparison is not strictly possible because there are no pairs of identical-aged poles from the two cratons. Nevertheless, the following two examples illustrate the method and the uncertainties of the present data set.

In Fig. 10*b*, the great-circle distance between the 2.23 Ma Malley pole (or the nearby 2.19 Ga Dogrib virtual geomagnetic pole) and the 2.03 Ga Lac de Gras pole of the Slave craton is similar to the great-circle distance between the 2.22 Ga Nipissing pole and either the 2.07 Ga Lac Esprit or 2.00 Ga Minto pole of the Superior craton, which might suggest that the two cratons could have been on the same plate at great distance from one another during this ~200 million years period. However, the pole ages for the two cratons are not matched, and there are no Slave poles *within* this long interval, so that further data are required before a definitive interpretation is possible. On the other hand, the great-circle distance between the Superior 2.22 Ga Nipissing and 2.17 Ga Biscotasing poles of Fig. 10*a* is substantially longer that the great-circle distance between the 2.23 Ga Malley and 2.19 Ga Dogrib poles, which might suggest that the two cratons were not on the same plate during this ~50 million years interval. However, once again, the time intervals for the two cratons are not matched, so that if the Superior polar wander occurred mostly in the latter half of the 2.22–2.17 Ga interval, it would go unrecorded in the older Slave data. In addition, the Dogrib virtual geomagnetic pole, based on only two dykes, is very poorly defined. Once again, further data are required to reach a definite conclusion.

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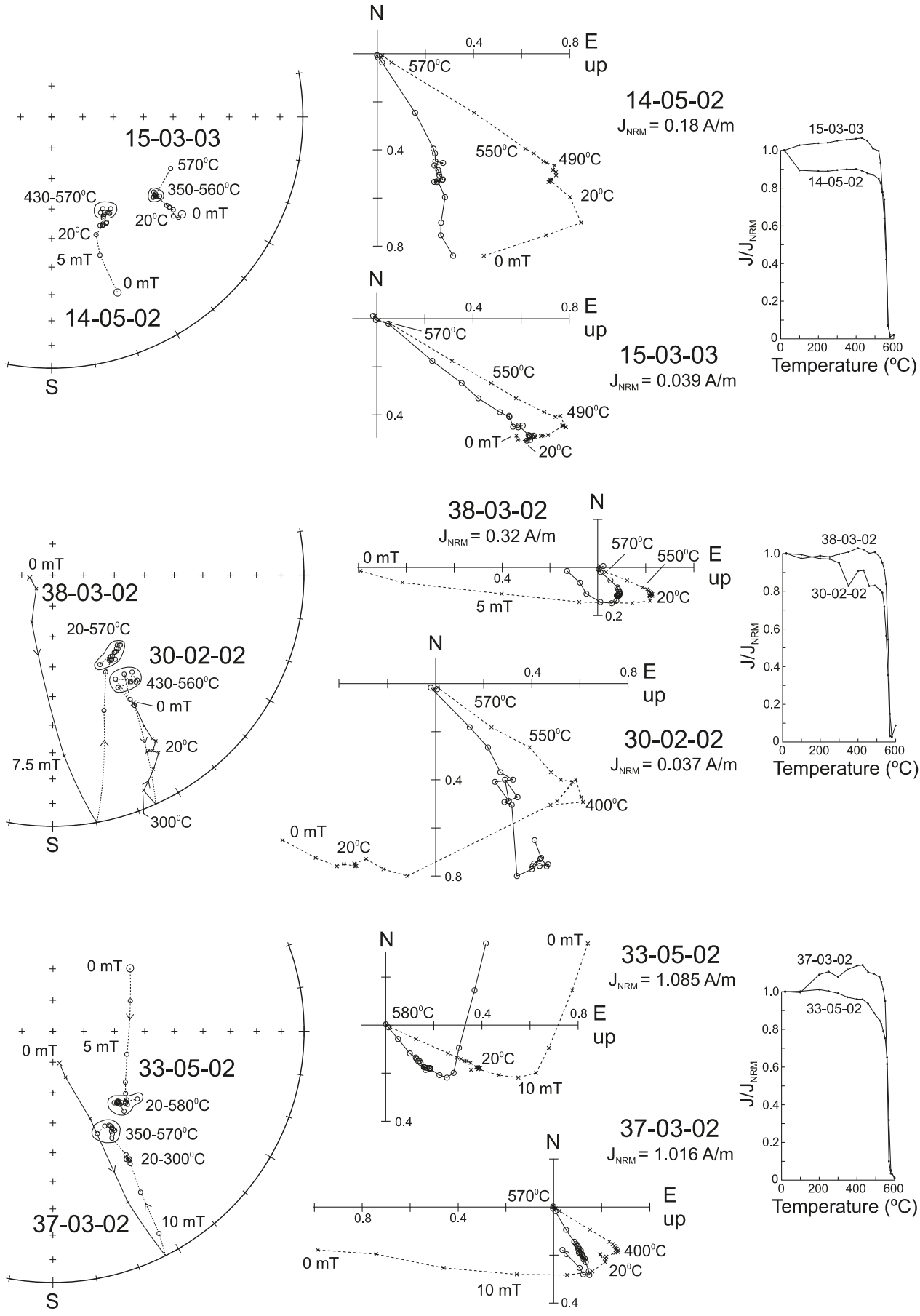
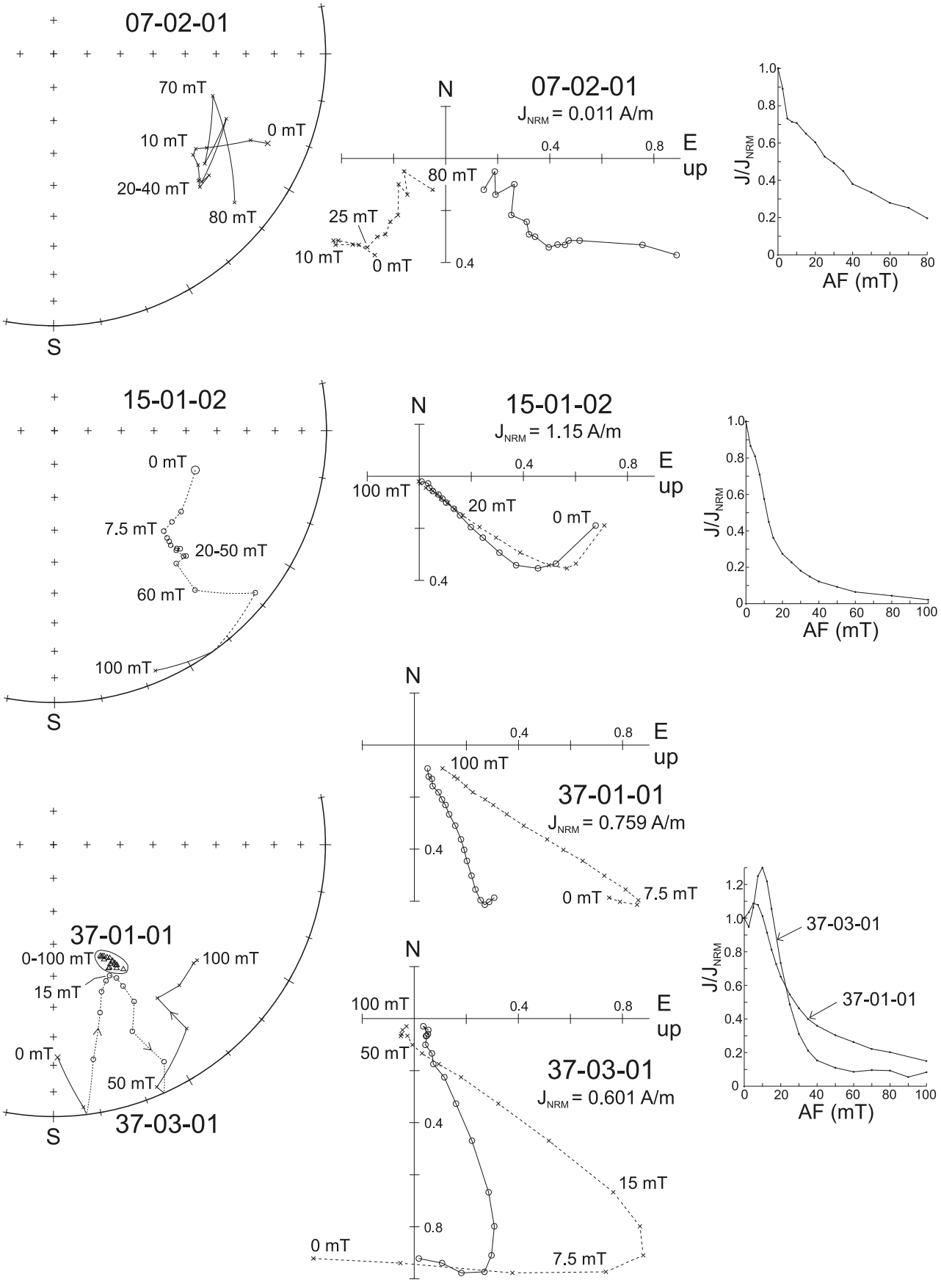
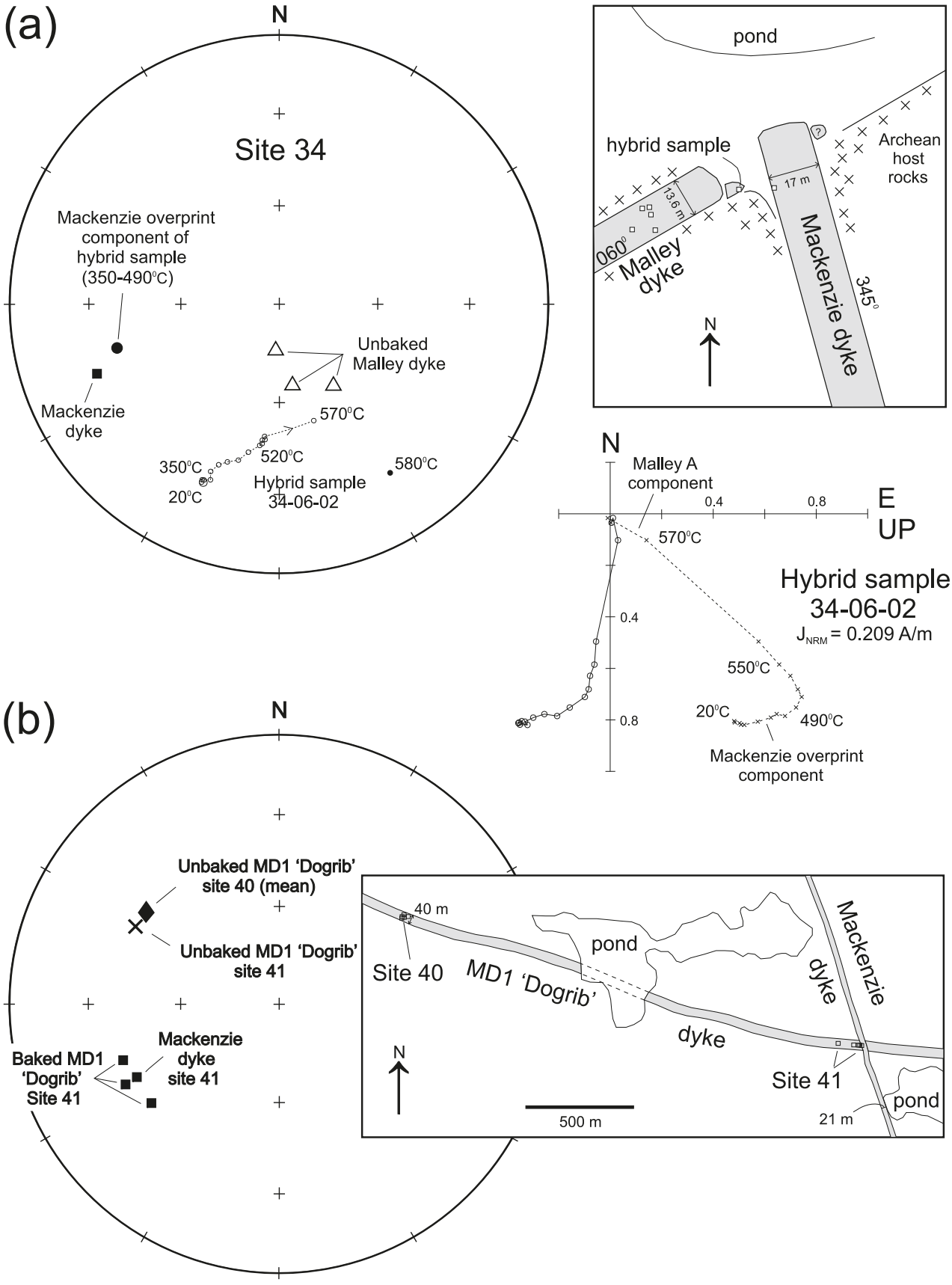


Fig. 8. Examples of paleomagnetic data for Malley dykes on AF demagnetization. Symbols as in Fig. 7.



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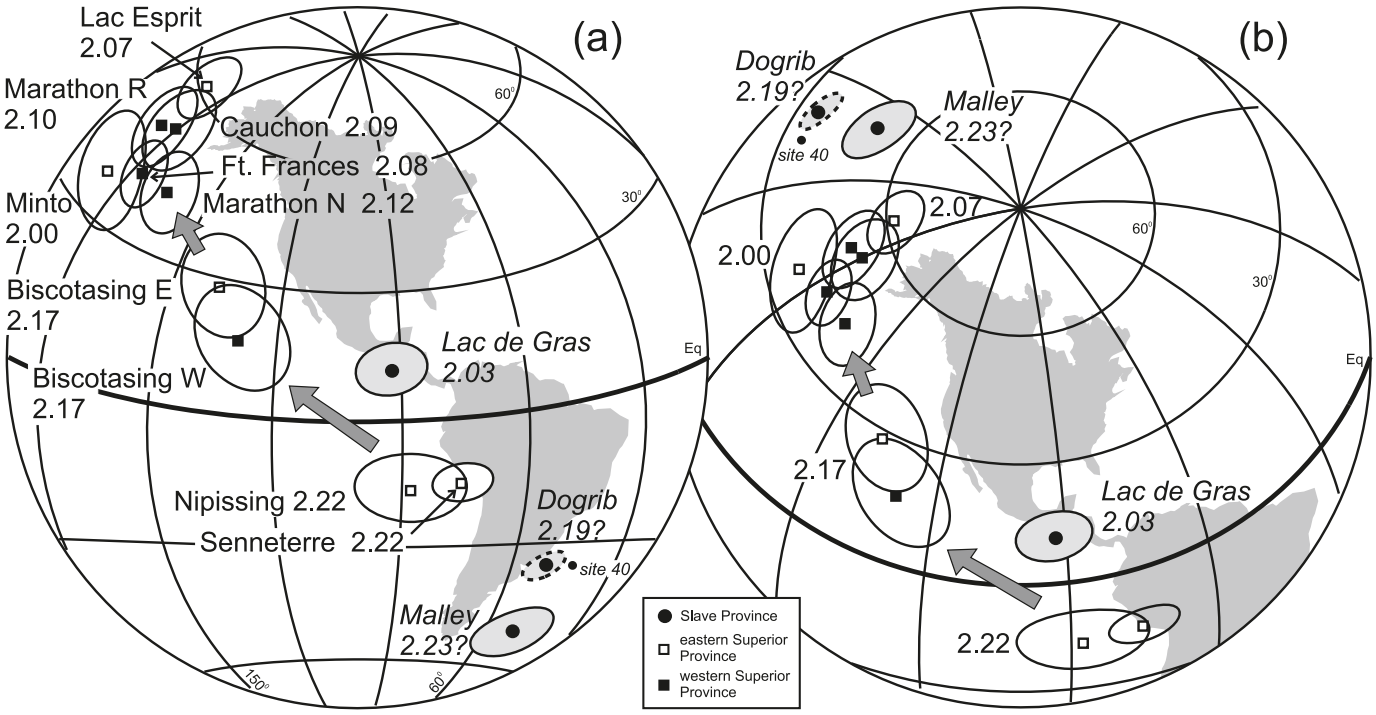
Fig. 9. Baked contact tests for (*a*) Mackenzie–Malley dyke intersection at site 34, and (*b*) Mackenzie – MD1 "Dogrib" dyke intersection at site 41. Symbols as in Fig. 7.



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Fig. 10. Comparison of Slave (circles) and Superior (squares) craton paleopoles. Pole ages are in Ga. Ellipses of 95% confidence are shown about each pole. The Malley dyke pole is from this study. The Dogrib dyke pole (McGlynn and Irving 1975) is a virtual geomagnetic pole that does not average out secular variation because it is based on only two dykes. Hence, its confidence ellipse (dashed) does not reflect the overall uncertainty in the pole. A small circle locates the virtual geomagnetic pole for site 40 of the MD1 "Dogrib" dyke in this study. Question marks for the Malley and Dogrib ages indicate that the poles for these units have not been demonstrated primary (see text). Other poles are well dated. Because of the long time interval between the Dogrib and Lac de Gras poles, their relative polarity is uncertain. The Malley and Dogrib poles are shown in (*a*) using the polarity option that minimizes their distance from the Lac de Gras pole, and in (*b*) using the alternative polarity option (see text). Superior paleopoles are only fully labelled in (*a*). Paleopoles from eastern (open squares) and western Superior (closed squares) are distinguished because they have been interpreted to indicate post-2.07 Ga relative rotation of the two portions of the craton (Halls and Davis 2004; Buchan et al. 2007*a*; Evans and Halls 2010). References for poles and ages are as follows: Lac de Gras dykes (Buchan et al. 2009); Senneterre dykes (Buchan et al. 1993, 1996); Nipissing N1 sills (Andrews et al. 1986; Noble and Lightfoot 1992; grand mean pole calculated in Buchan et al. 1998); Biscotasing dykes (Buchan et al. 1993; Halls and Davis 2004); Marathon Normal (N) and Reversed (R) dykes (Buchan et al. 1996; Hamilton et al. 2002; Halls et al. 2008; Evans and Halls 2010); Cauchon Lake dykes (Halls and Heaman 2000); Fort Frances dykes (Halls 1986; Buchan et al. 1996); Lac Esprit dykes (Buchan et al. 2007*a*); Minto dykes (Buchan et al. 1998).

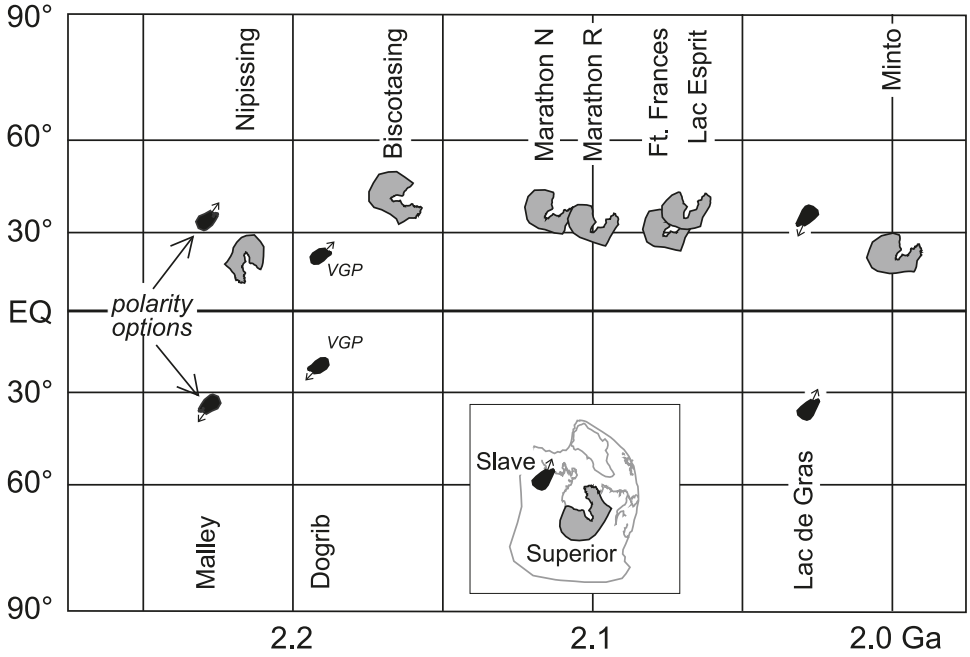


Conclusions

The main conclusions of this paper are as follows:

1. Northeast-trending Malley dykes of the east-central Slave craton were emplaced at 2231 ± 2 Ma based on U–Pb baddeleyite dating.
2. The Malley dyke swarm may continue northeast across the Bathurst fault as the Brichta swarm, for a total swarm length of ~500 km, based on a recontruction of the eastern Slave craton, which restores 115 km of left-lateral offset along the fault (Tirrul and Grotzinger 1990), as well as strong geochemical similarities between Malley and Brichta dykes.
3. Malley dykes, including the dated dyke, carry a stable high unblocking temperature characteristic remanence, with a mean direction of D = 138.3°, I = –53.8° (α_{95} = 5.8°, k = 81), and a corresponding paleopole is at 50.8°S, 50.0°W (dm = 8.1°, dp = 5.6°).
4. The Malley remanence was acquired before 2.03 Ga, and likely before 2.19 Ga. However, a field test is required to establish if the remanence is primary.
5. The paleopoles from well-dated units in the Slave craton suggest that plate tectonic processes were operating within the 2.23–2.03 Ga period.
6. If the Malley paleopole is primary, a comparison between Slave craton paleopoles for the 2.23 Ga Malley dykes and 2.03 Ga Lac de Gras dykes with poles on the Superior craton APWP between 2.22 and 2.00 Ga indicates the following:
   a. The two cratons were not in their current relative position during this period.
   b. The two cratons likely did not drift in close proximity to one another as part of a single (super)continent throughout the period, in agreement with geological evidence.
   c. Further studies are needed to test whether they drifted together for part of this period or whether they may have drifted at a distance from one another on the same tectonic plate.

Fig. 11. Drift plot for Superior and Slave cratons in the 2.23–2.00 Ga interval. Paleolatitudes and azimuthal orientation are based on the mean paleopoles shown in Fig. 10. Longitude is unconstrained by the palemagnetic data. Because of paleomagnetic polarity ambiguity, when the Superior craton is placed in one hemisphere, the Slave craton is permitted to occupy a location in either hemisphere, as shown. The arrow on the Slave craton indicates present-day north. The Malley and Dogrib positions for the Slave craton are based on poles that have yet to be demonstrated as primary with a field test (see text). Positions for the Superior craton (shown as a whole) are based on data from units in both the eastern Superior (Nipissing, Biscotasing, Lac Esprit, and Minto) and western Superior (Marathon Normal (N), Marathon Reversed (R), and Ft. Frances). The Slave position at 2.19 Ga is labelled VGP because it is derived from a virtual geomagnetic pole (see text). The figure was created with the assistance of the GMAP32 computer program (Torsvik and Smethurst 1997). EQ, equator.



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